

Lost & Found Item Management System

Project Overview

The Lost & Found Item Management System is a console-based Java application that helps users report lost and found items and view them efficiently using a MySQL database.

Database Script

```
CREATE DATABASE lostfound;
USE lostfound;

CREATE TABLE item (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    place VARCHAR(50),
    status VARCHAR(10)
);
```

DTO – Item Class

```
package dto;

public class Item {
    private int id;
    private String name;
    private String place;
    private String status;

    public Item() {}

    public Item(String name, String place, String status) {
        this.name = name;
        this.place = place;
        this.status = status;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getPlace() { return place; }
    public void setPlace(String place) { this.place = place; }
    public String getStatus() { return status; }
    public void setStatus(String status) { this.status = status; }
}
```

DAO Interface

```
package dao;
import dto.Item;
```

```
import java.util.List;

public interface ItemDAO {
    void addItem(Item item);
    List<Item> viewItems(String status);
}
```

Database Connection Utility

```
package util;

import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/lostfound",
                "root",
                "password"
            );
        } catch (Exception e) {
            e.printStackTrace();
        }
        return con;
    }
}
```

DAO Implementation

```
package daoimpl;

import dao.ItemDAO;
import dto.Item;
import util.DBConnection;
import java.sql.*;
import java.util.*;

public class ItemDAOImpl implements ItemDAO {

    public void addItem(Item item) {
        try {
            Connection con = DBConnection.getConnection();
            String sql = "INSERT INTO item(name, place, status) VALUES (?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, item.getName());
            ps.setString(2, item.getPlace());
            ps.setString(3, item.getStatus());
            ps.executeUpdate();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```

    }

    public List<Item> viewItems(String status) {
        List<Item> list = new ArrayList<>();
        try {
            Connection con = DBConnection.getConnection();
            String sql = "SELECT * FROM item WHERE status=?";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, status);
            ResultSet rs = ps.executeQuery();

            while (rs.next()) {
                Item i = new Item();
                i.setId(rs.getInt("id"));
                i.setName(rs.getString("name"));
                i.setPlace(rs.getString("place"));
                i.setStatus(rs.getString("status"));
                list.add(i);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return list;
    }
}

```

Main Application

```

package main;

import dao.ItemDAO;
import daoimpl.ItemDAOImpl;
import dto.Item;
import java.util.*;

public class MainApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ItemDAO dao = new ItemDAOImpl();

        while (true) {
            System.out.println("\n--- LOST & FOUND SYSTEM ---");
            System.out.println("1. Report Lost Item");
            System.out.println("2. Report Found Item");
            System.out.println("3. View Lost Items");
            System.out.println("4. View Found Items");
            System.out.println("5. Exit");

            int ch = sc.nextInt();
            sc.nextLine();

            if (ch == 1 || ch == 2) {
                System.out.print("Item Name: ");
                String name = sc.nextLine();
                System.out.print("Place: ");
                String place = sc.nextLine();
            }
        }
    }
}

```

```
        String status = (ch == 1) ? "LOST" : "FOUND";
        dao.addItem(new Item(name, place, status));
        System.out.println("Item added successfully!");
    }
    else if (ch == 3 || ch == 4) {
        String status = (ch == 3) ? "LOST" : "FOUND";
        List<Item> items = dao.viewItems(status);
        for (Item i : items) {
            System.out.println(i.getId()+" "+i.getName()+" "+i.getPlace()+" "+i.getStatus());
        }
    }
    else if (ch == 5) {
        System.out.println("Thank you!");
        break;
    }
}
sc.close();
}
```